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Primate Locomotory Adaptations

Arboreal and Terrestrial

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Why study locomotion?

- Man's full time, bipedal, upright locomotion is unique among the primates and its origins are unclear from the fossil record
- Further, most of evolutionary developments cannot be fully understood by studying skeletons alone
- A major contribution of studying living primates is functional anatomy
- Two functional complexes that have received maximum attention are hands and feet – not very different as except man, every other primate uses both for locomotion

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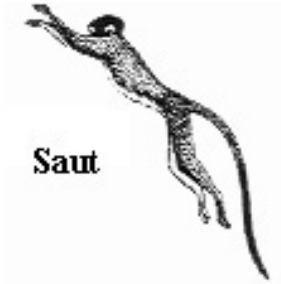
Types of Adaptations

Four Types

1. Vertical clinging
2. Quadrupedalism
3. Brachiation
4. Bipedalism



Marche sur les branches



Saut



Bipédie



Suspension



Quadrupédie
avec knukle-walking



Quadrupédie

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Vertical Clinging & Leaping

- Found among **Prosimians**
- Leaping from one vertical support to another, using only hind limbs for propulsions
- Forelimbs are used for clinging to trunks

Adaptations

- Elongation of hind limbs - used as springs in leaping

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Vertical Clinging & Leaping



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Vertical Clinging & Leaping



Indri or Babakoto
Indri indri

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Quadrupedalism

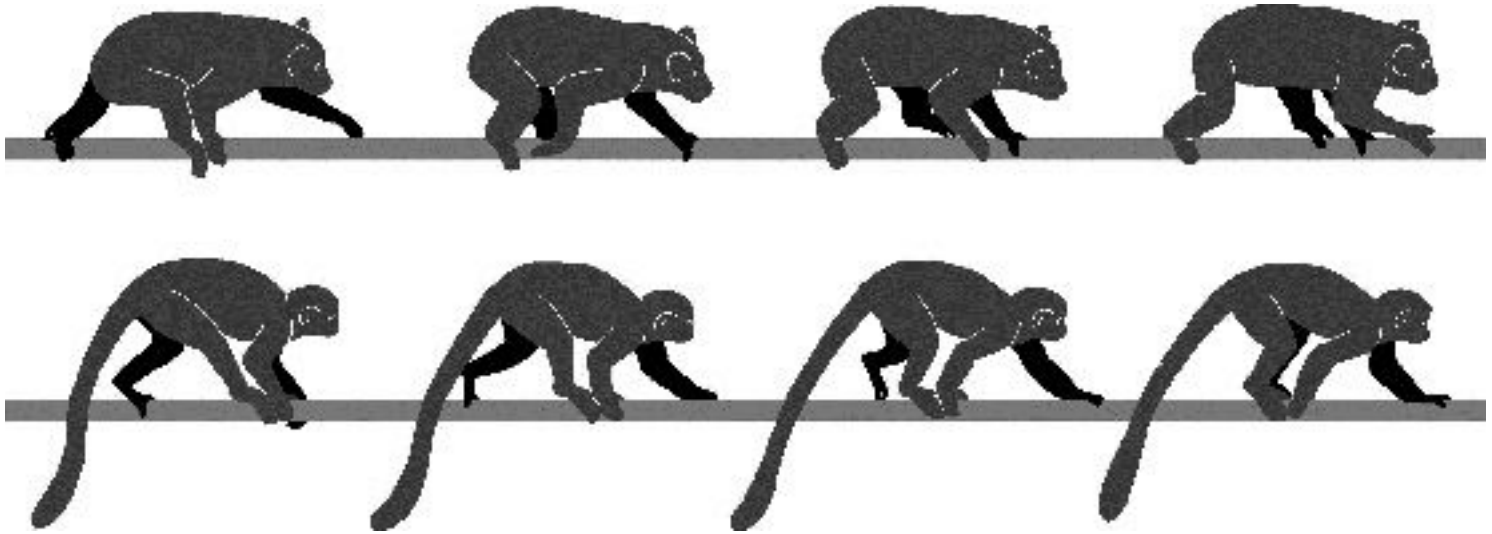
- Involves both fore and hind limbs
- Two sub-types
  - **Arboreal Quadrupedalism**
  - **Terrestrial Quadrupedalism**

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Arboreal Quadrupedalism

- **Old and New World Monkeys**



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Arboreal Quadrupedalism

Adaptations

1. Reduction in index finger / thumb
2. Big toe splayed out at a wide angle - forceps to enclose a branch
3. Short and broad trunks - COG
4. Fore and hind limbs are of equal length
5. Prehensile hands & feet - to grasp branches
6. N.W.M. - Long Prehensile tails

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Terrestrial Quadrupedalism

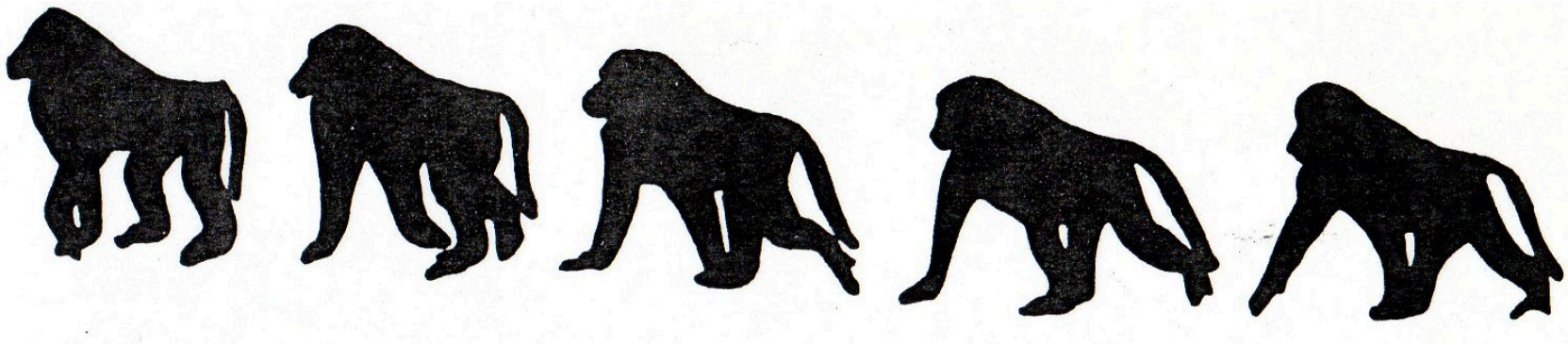
- Two Sub-types
  - Ground Running – Baboons
  - Knuckle Walking – Chimpanzees & Gorillas

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Ground Running

- Walking on soles of their feet
- Limbs of equal length
- All fingers of equal length - allows more precision grip

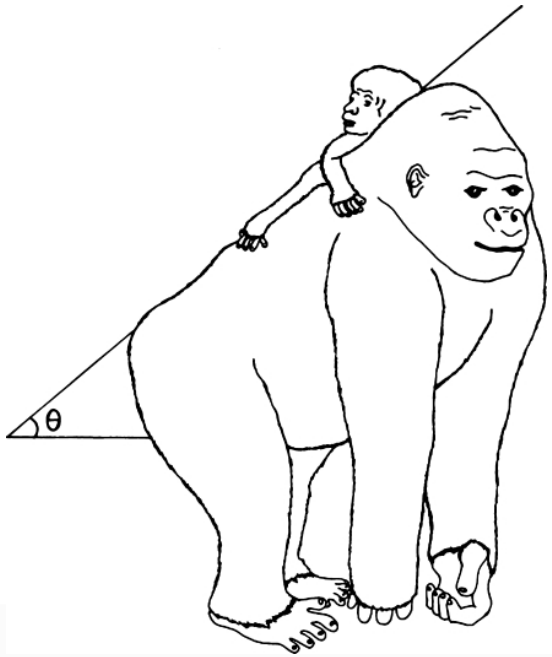


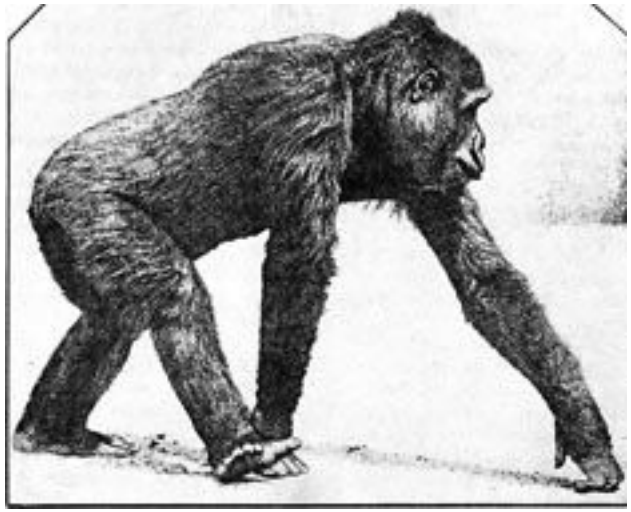
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Knuckle Walking

- **Chimp** and **Gorilla**
- Adaptation because of large body size but at the same time retaining arboreal features like long arms and long fingers and short thumbs like their brachiating ancestors



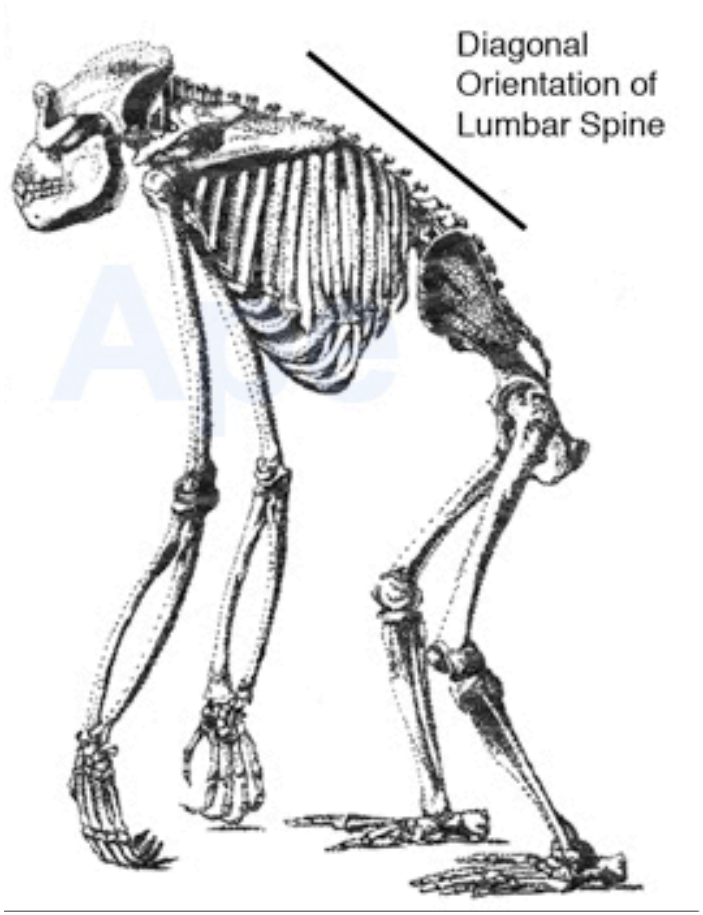


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Knuckle Walking

1. Elongation of pelvis
2. Head of femur - not forming an angle with shaft - restricts the movement of the leg
3. Habitually bent knee
4. Whole foot making contact with surface



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Knuckle Walking





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Brachiation

- **Gibbon / Siamang / Orangutan**
- Progression is accompanied by swinging the body from one handhold to another
- Primary adaptation for feeding at terminal ends of branches



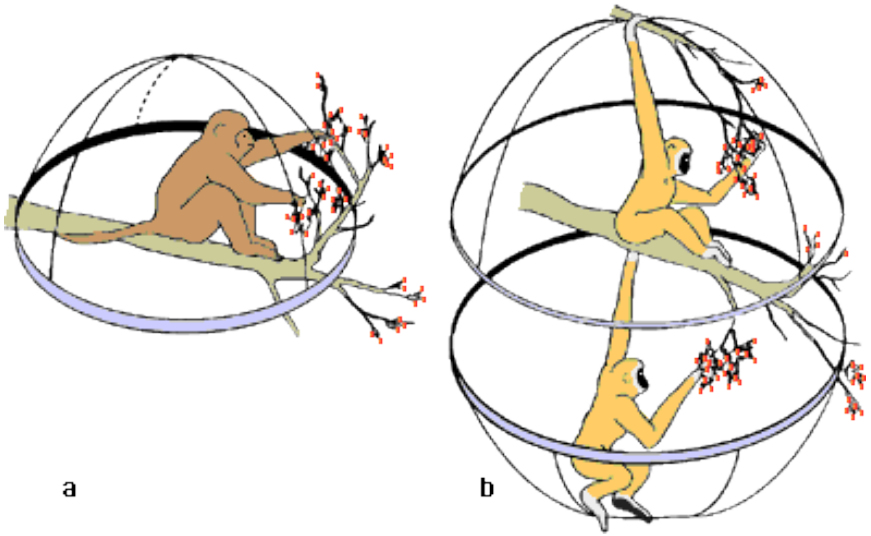
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True Brachiation

Adaptations:

1. Relatively light body weight
2. Long arms with long fingered hands
3. Long opposable thumb is in the way and hence all brachiators have some means of getting their thumbs out of way during locomotion - Gibbon tucks away the thumb - since hands function as hooks



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Gibbon's True Brachiation



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Modified Brachiation

- Found in Orangutan
- Involves arm-swinging with feet used for support, climbing and hanging.
- The orangutans are really cautious climbers than brachiators, and climb using any combination of their four grasping extremities.



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Group	Hand Movement	Thumb	Grasping Patterns
Tupaioidea	Convergence and divergence only	Non-opposable	Use two-handed prehension, whole-hand control
Lemuroidea Lorisoidea Tarsioidea	Convergence and prehension	Pseudo-opposable	Use a single grip pattern for power and precision, whole-hand control
Ceboidea	Convergence and prehension	Pseudo-opposable	Some separation of precision and power grips, but precision grip different from Cercopithecoid-Hominoid type. Little evidence of independent finger control.
Cercopithecoidea	Convergence and prehension	Truly opposable	Use separate power and precision grips. Have separate control of the thumb and, in some species, separate control of the index finger as well.
Hominoidea	Convergence and preshension	Truly opposable	Use separate power and precision grips. Have separate control of all fingers.

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Bipedalism

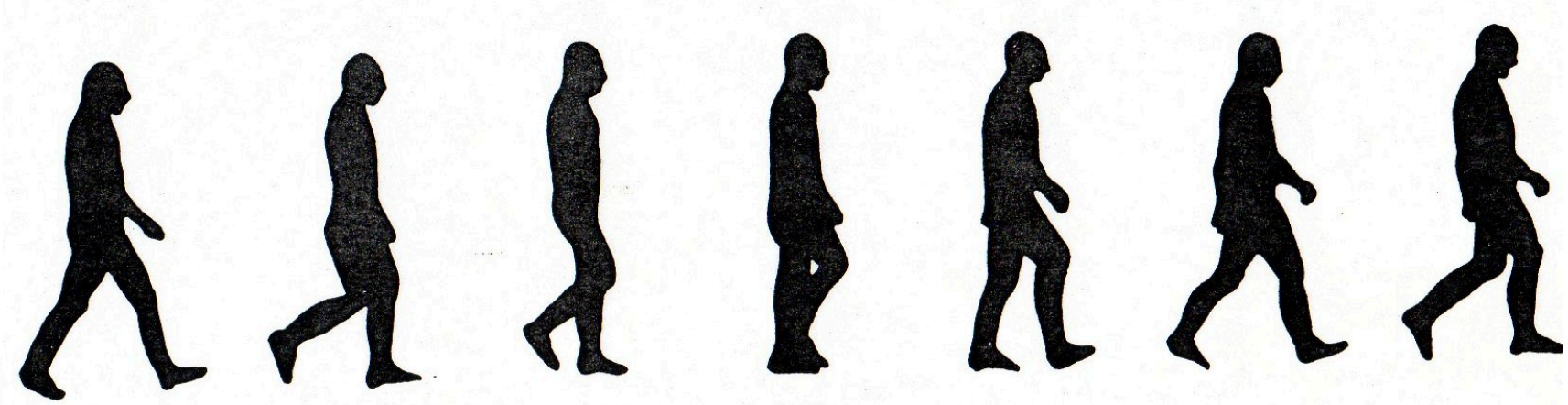
Associated Changes

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Bipedalism

- Hind limbs alone support the body weight and propel it along the ground
- The reasons for this adaptation are unknown and speculative
- Other primates have come to the ground but none taken to Bipedalism



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Bipedalism

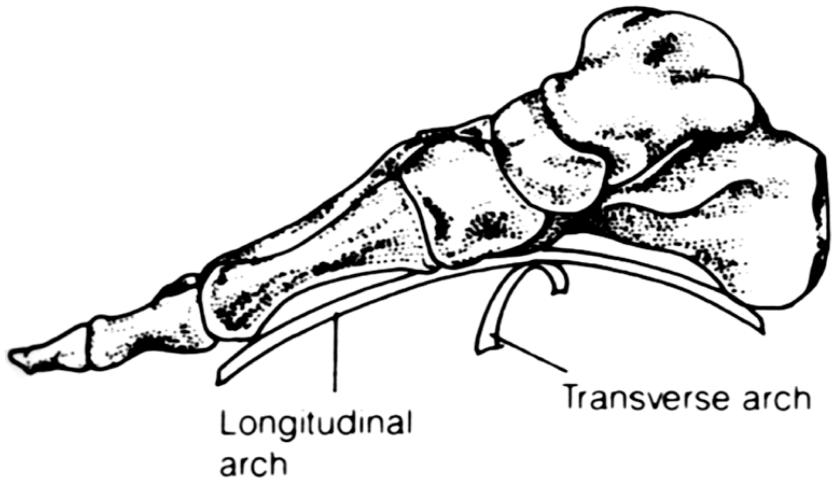
- Evolved some four million years ago
- Changes to support bipedalism in human anatomy involved a number of structures – head, foot, pelvis, knee, leg, vertebral column, musculature etc.
- Though the tendency to occasional bipedalism can be seen in other primates, the structural adaptations seen in humans is extraordinary
- A number of theories speculate the origin of this locomotor adaptation – more later

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The Human Foot

- Redesigned as a platform to support entire body weight rather than a grasping structure
- Reduction in size of toes
- Development of two arches to support weight
- Axis of the human foot lies between big toe & first toe

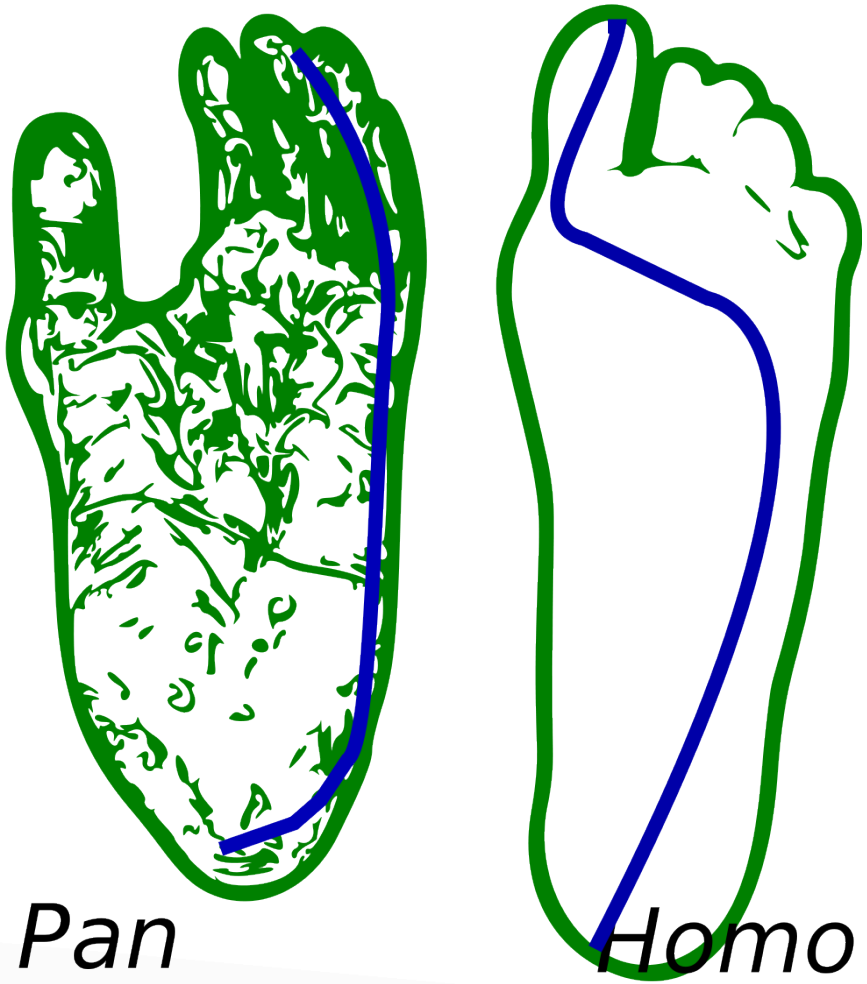


(*b*) Human (medial view)

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Human Foot



- Transference of weight from the heel, along the outside of the foot, across the ball of the foot, and finally through the big toe – conserves energy during locomotion

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Human Foot



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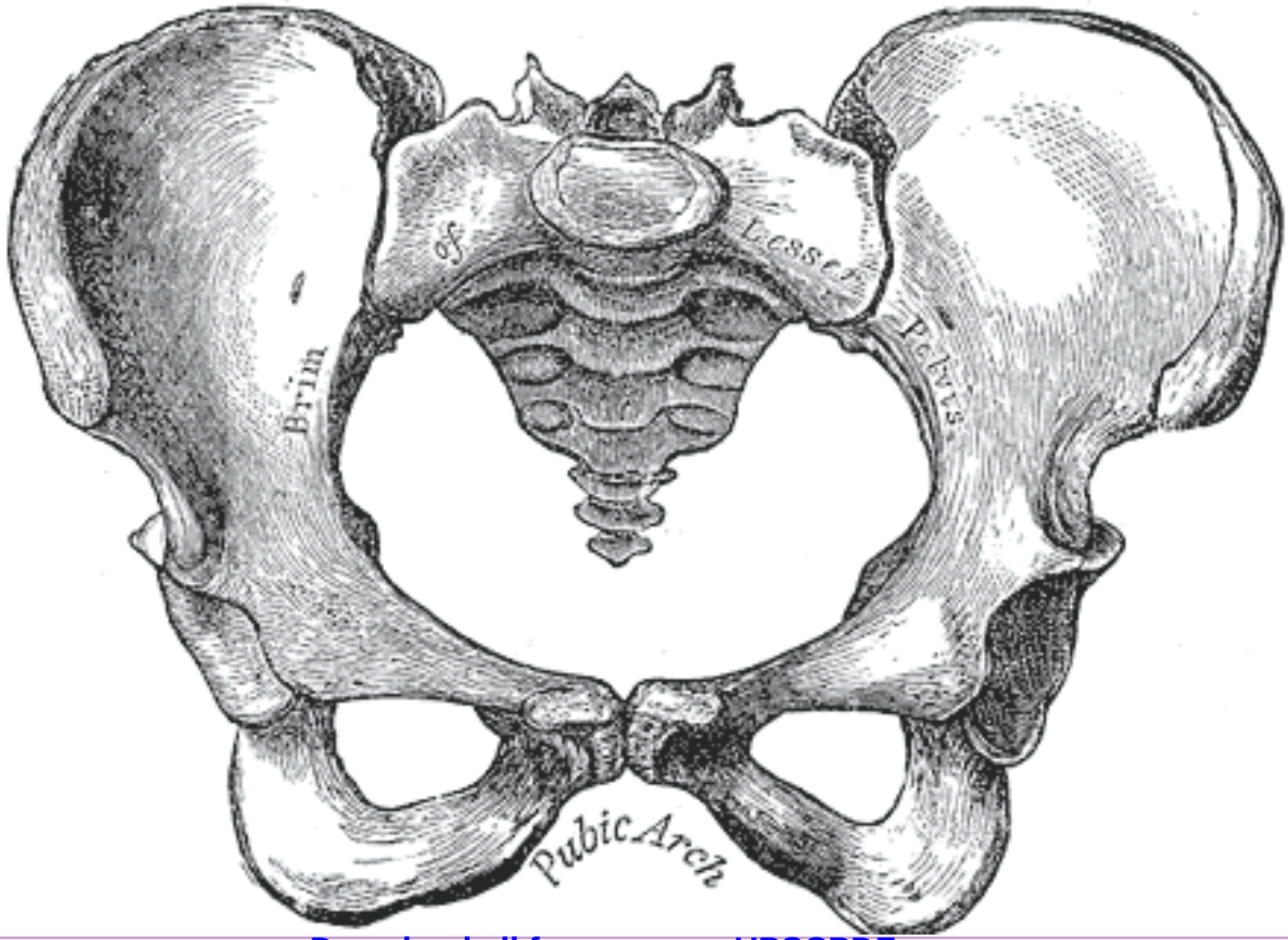
Hip

- Shorter and broader than quadrupedal ancestors - to support more weight
- Pelvic region has become basin-shaped to support vertical weight-transference
- Three muscles, **gluteus maximus, gluteus medius** and **gluteus minimus** stabilize pelvis
- Ischium, a pelvic bone, flattened allowing humans to sit comformtably

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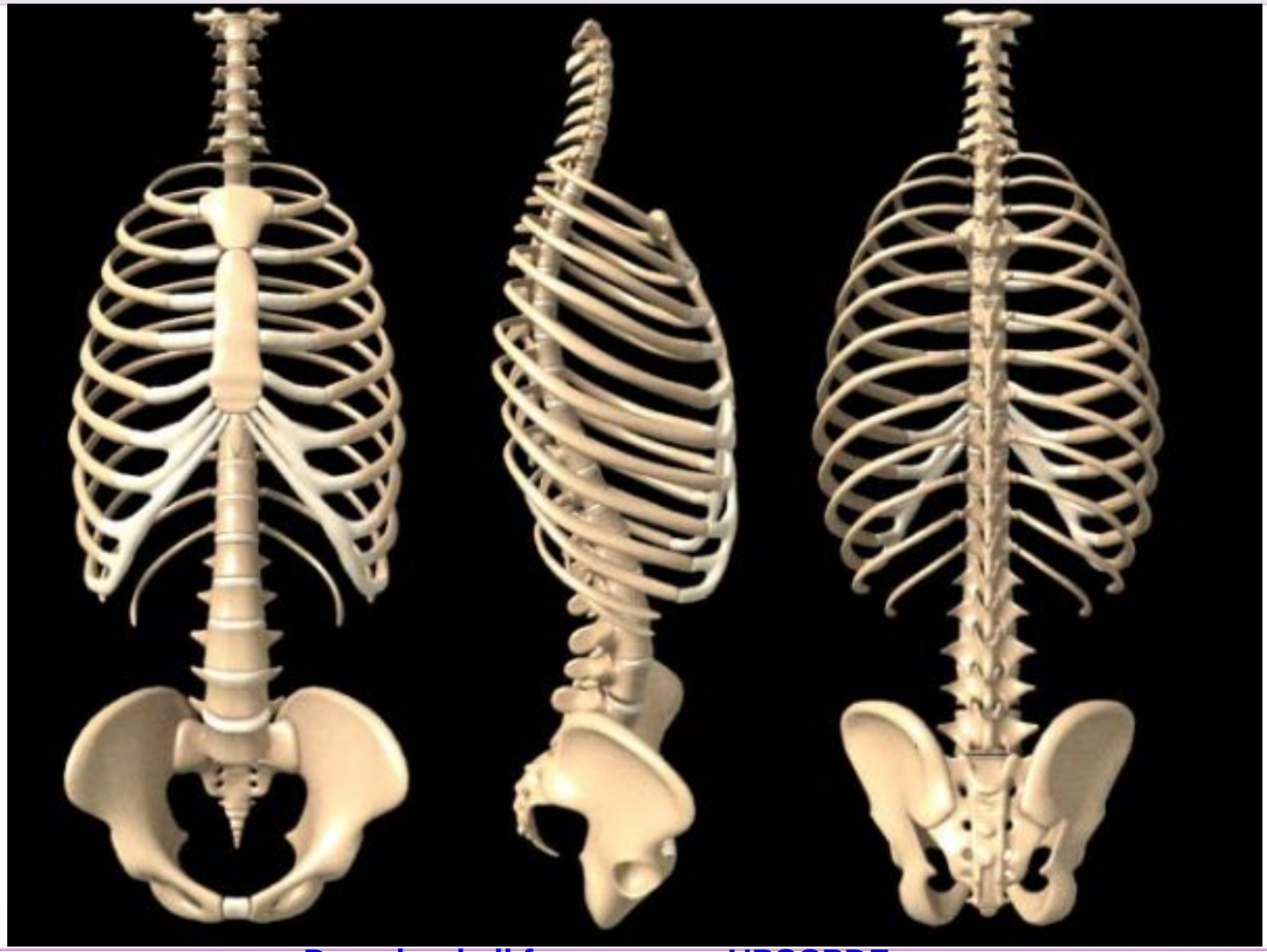
Hip



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Hip

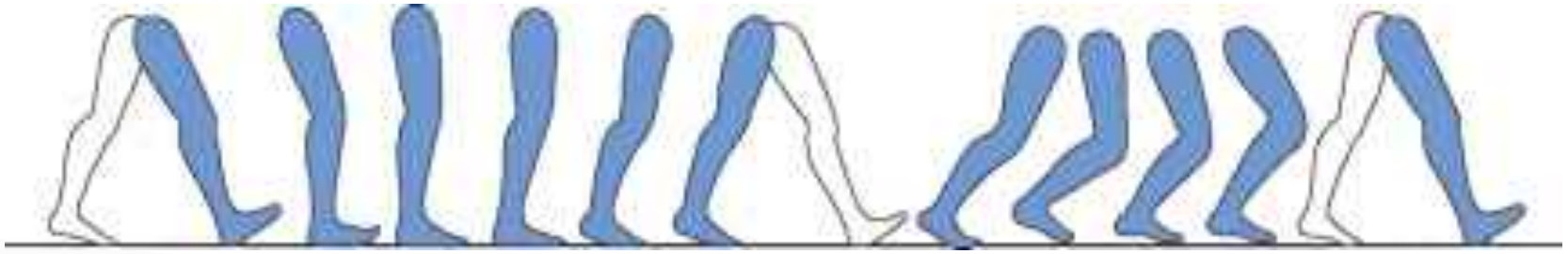


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Knee

- Knee joints have enlarged to support increased body weight
- Degree of knee extension – angle between thigh and shank reduced (hence no bent knee posture)
- **Double Knee Action** reduces energy during walking – when we walk, the thigh is bent inwards and knees straight so that knees are almost directly under the body, rather than out to side

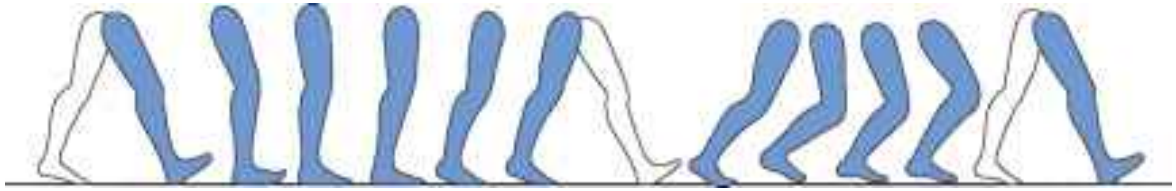


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Changes in the Leg

- Leg and hip musculature highly developed to facilitate upright standing, walking, running and climbing
- Increase in the length of leg – allowing natural swing of leg during walking
- Forelimbs freed from locomotion – hence used for manipulation of objects
- Thigh muscles **sartorius, rectus femoris** and **biceps femoris** and calf muscles **gastrocnemias** are well developed



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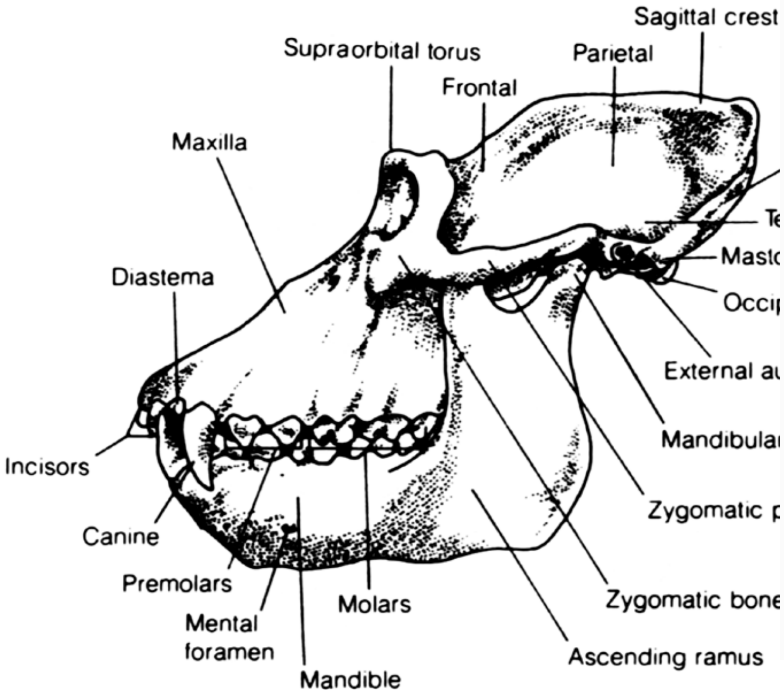
Skull

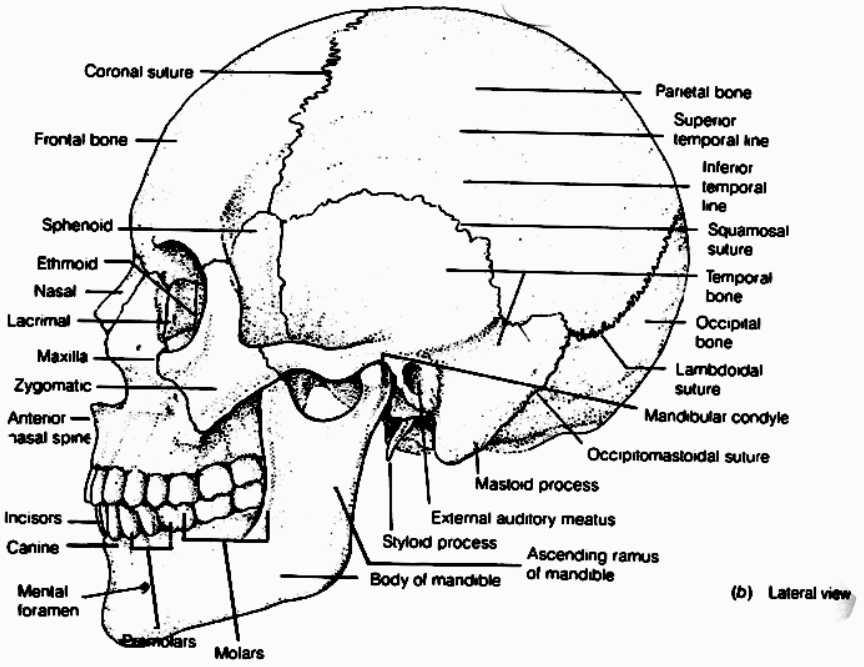
- Skull balanced on vertebral column
- Foramen magnum located inferiorly, which puts weight of the head behind the spine
- Flat human face helps balancing on two occipital condyles
- Thus, without strong supraorbital ridges and muscles of forehead (occipitofrontalis), we are able to balance the head
- We use these muscles only for facial expressions

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Skull



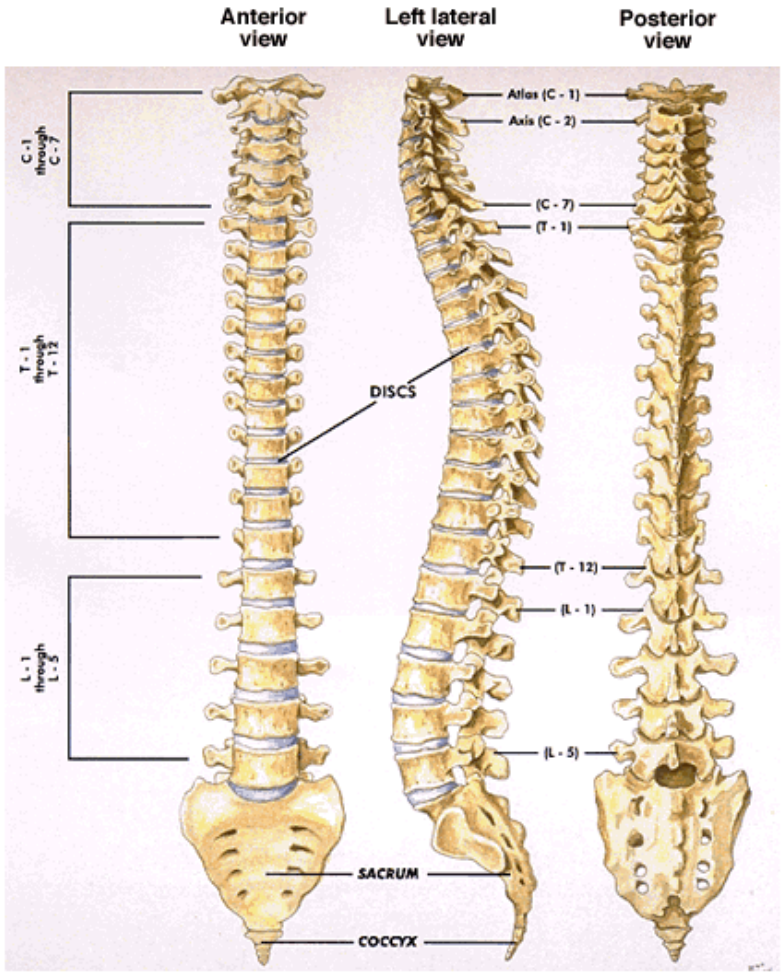


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Vertebral Column

- The vertebral column of humans takes a forward bend in the lumbar (lower) region and a backward bend in the thoracic (upper) region.
- Typical S-Curve
- The curves help conserving energy while also help transferring the weight and CoG directly over the feet

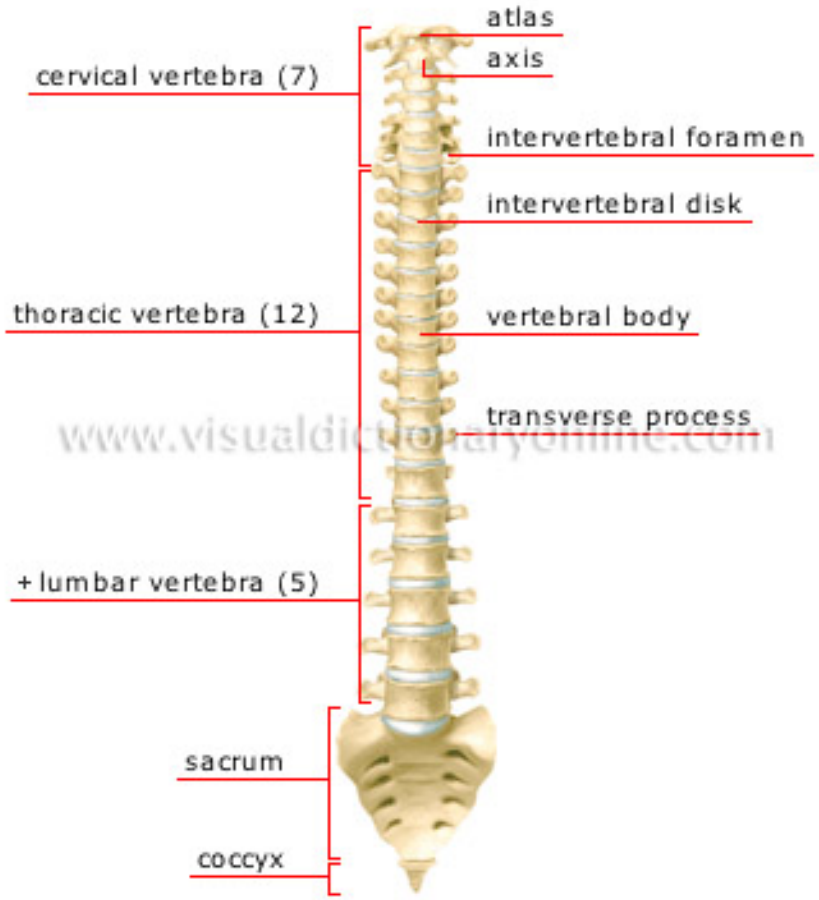


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Vertebral Column

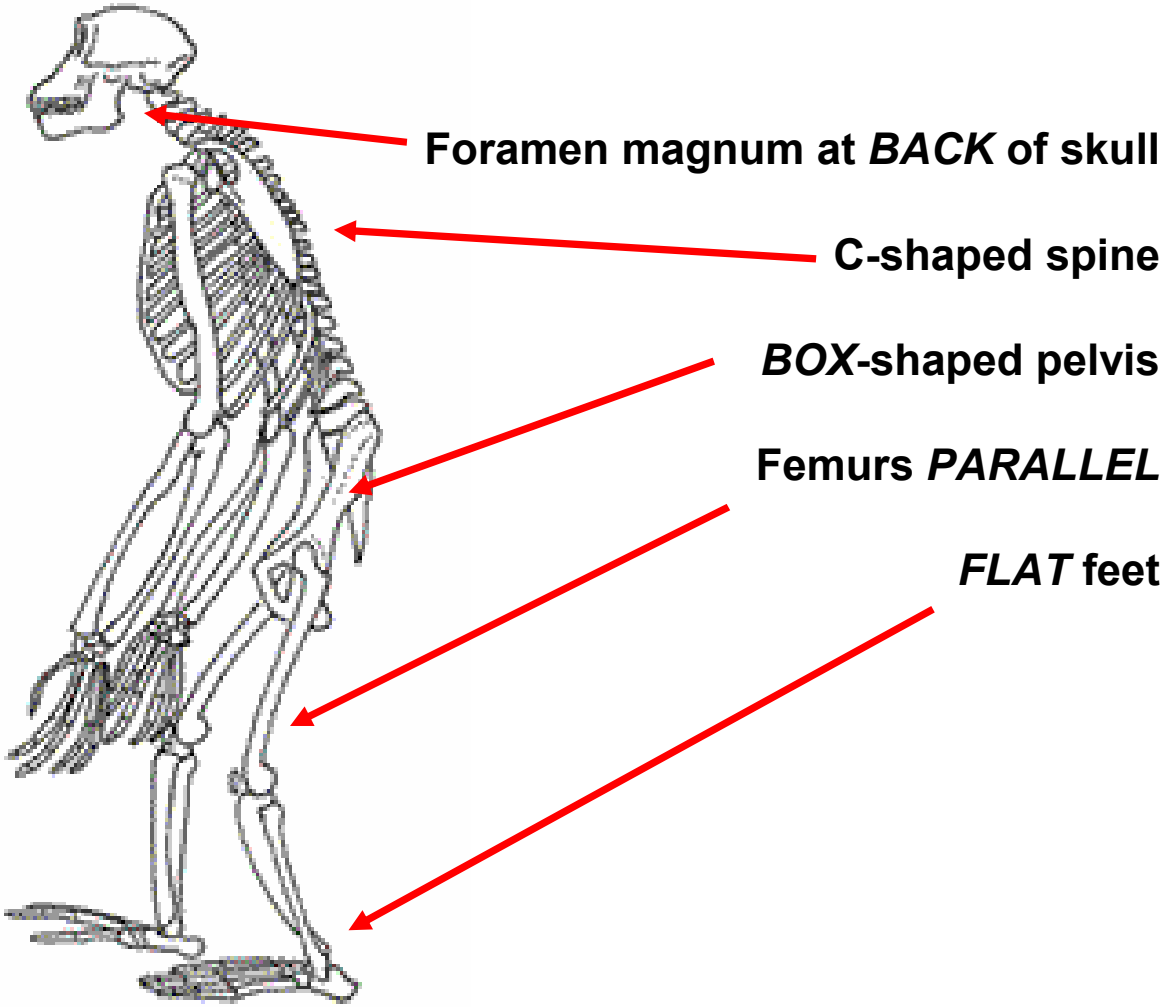
- Also, the degree of body erection (the angle of body incline to a vertical line in a walking cycle) is significantly smaller to conserve energy.
- Fusion of Sacral Vertebrae
- Vertebrae from top to bottom gradually increase in size for better load bearing giving human column its characteristic cone shape

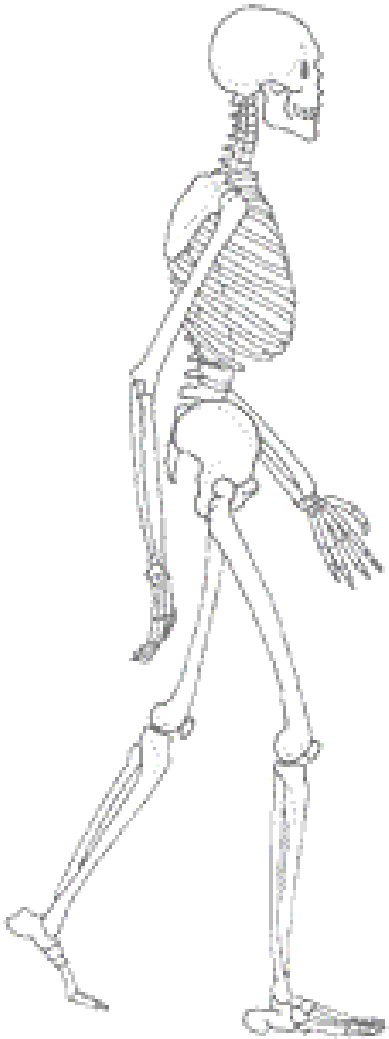


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Quadrupedal ape skeletons:





Source: http://www.amonline.net.au/human_evolution/skeleton/more_info.htm

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Adaptations for bipedalism:



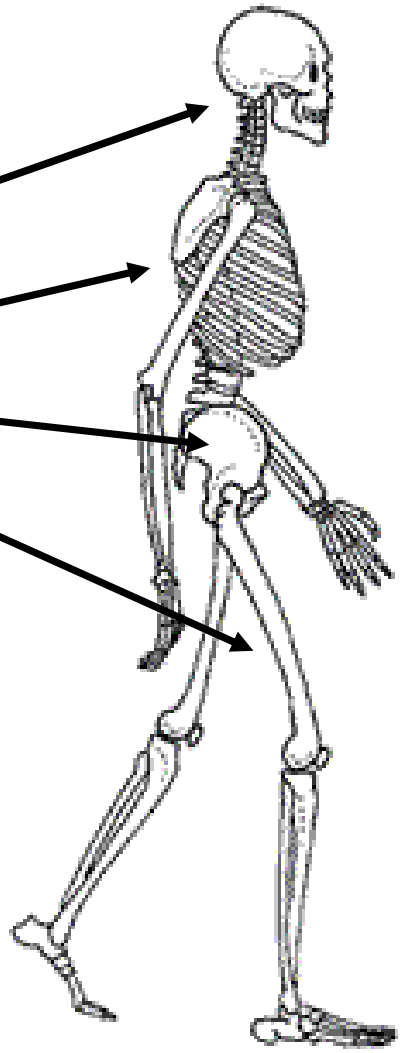
Foramen magnum at *BASE* of skull

S-shaped spine

***BOWL*-shaped pelvis**

Femurs *SLANT INWARDS* to knees

***ARCHED* feet**

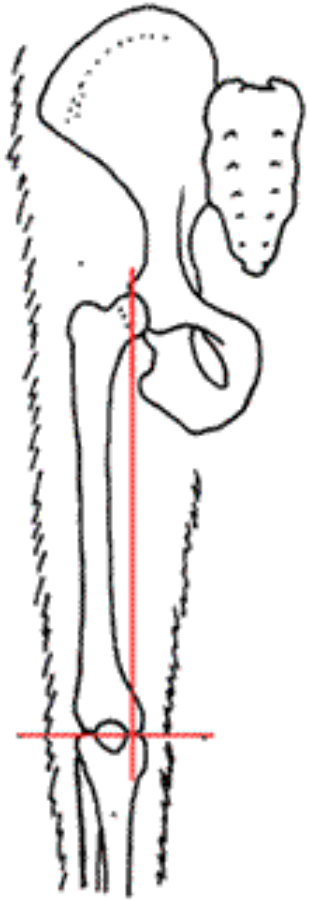


Source: http://www.amonline.net.au/human_evolution/skeleton/more_info.htm

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Femur orientation:



Chimpanzee



Modern Human

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Why Bipedalism?

- What factors have resulted in this very crucial development in hominid evolution?
  - Tool Use – Darwin & Washburn
  - Energy Efficiency – Isbell / Young
  - Body Temperature – Wheeler
  - Habitat Variability – Potts
  - Reproduction – Lovejoy
  - Canine Reduction - Jolly

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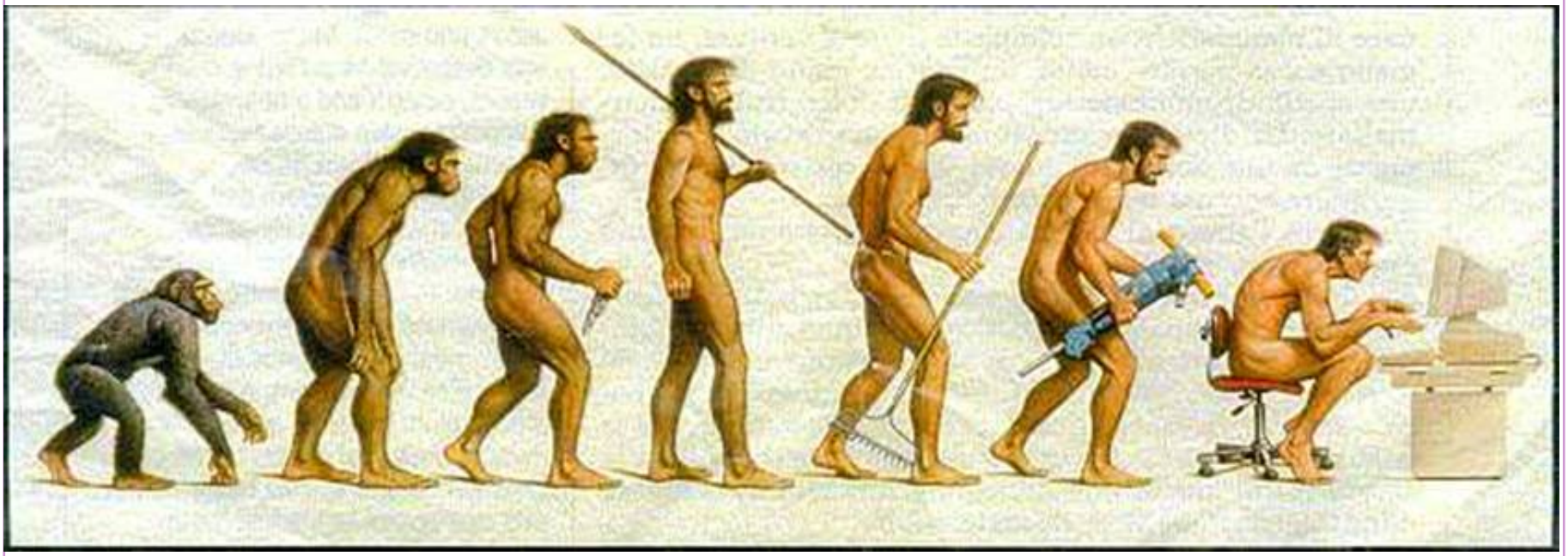
Evolutionary consequences of bipedal functional anatomy

- Some features of humans are still poorly adapted to bipedalism leading to negative implications
- Lower back and knee are susceptible to osteological malfunction
- This is because the human body weight is now supported by joints of two legs rather than four legs
- Vertebral arthritis was discovered in paleolithic hunter gatherers too – shows its not a civilization problem

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Where are we headed?



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